



**Software Project Management
Assignment-6**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which one of the following charts would be the most useful to decompose the project activities into smaller tasks that are more meaningfully managed?

- a) PERT chart
- b) GANTT chart
- c) Task network representation
- d) Work breakdown structure
- e) Pareto chart

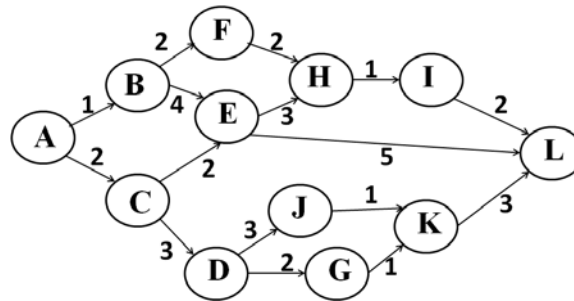
Correct Answer: d) Work breakdown structure

Detailed Solution:

A Work Breakdown Structure (WBS) is specifically designed to decompose a project into smaller, manageable tasks, making it ideal for organizing and managing project activities effectively.

QUESTION 2:

Which one of the following is the critical path for the activity network given below?



- a) A, B, E, H, I, L
- b) A, B, F, H, I, L
- c) A, C, D, J, K, L
- d) A, C, D, G, K, L
- e) A, B, F, H, I, L

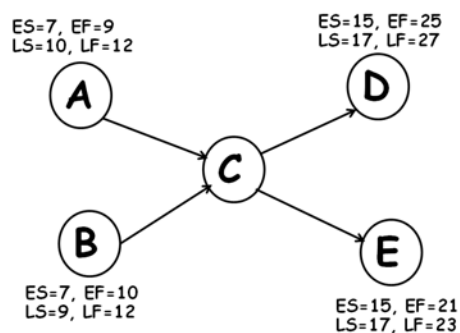
Correct Answer: c) A, C, D, J, K, L

Detailed Solution:

The path with the longest duration is the critical path.

QUESTION 3:

Consider a portion of a PERT network diagram given below. What is the LF of activity C?



- a) 10
- b) 11
- c) 16
- d) 17
- e) 20

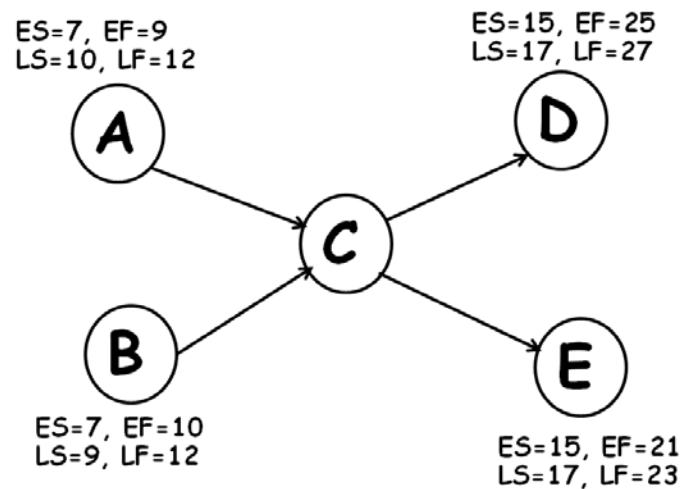
Correct Answer: d) 17

Detailed Solution:

The LF of activity C is the minimum LS of all its immediate successors, which is 17 in this case.

QUESTION 4:

Consider a portion of a PERT network diagram given below. What is the ES of activity C?



- a) 9
- b) 10
- c) 11
- d) 14

Correct Answer: b) 10

Detailed Solution:

The Early Start of activity C is the latest finish time of its predecessors, which is 10.



QUESTION 5:

PERT method differs from CPM in which one of the following aspects?

- a) PERT uses statistical time durations, whereas CPM uses deterministic time durations.
- b) PERT uses dummy activities, whereas CPM does not.
- c) PERT uses free float, whereas CPM uses total float in critical path calculations.
- d) PERT uses activity-on-arc, whereas CPM uses activity-on-node networks.
- e) PERT is primarily used for research and development projects, whereas CPM is used for construction and production projects.
- f) PERT focuses more on time, whereas CPM also considers cost optimization.
- g) PERT is event-oriented, whereas CPM is activity-oriented.

Correct Answer: a) PERT uses statistical time durations, whereas CPM uses deterministic time durations.

Detailed Solution:

PERT is a probabilistic, time-focused technique using three time estimates (optimistic, most likely, pessimistic), while CPM uses fixed, deterministic durations and focuses on cost and time trade-offs.

QUESTION 6:

Consider the data regarding the activities, their duration, and precedence for a certain project given in the following table. What is the total project duration?

Activity	Duration	Predecessors
A	14	-
B	13	A
C	20	A
D	10	B,C

a) 37

b) 44

c) 47

d) 57

e) 60

Correct Answer: b) 44

Detailed Solution:

Activity A starts at time 0 → Finishes at $0 + 14 = 14$

Activity B starts after A → Starts at 14 → Finishes at $14 + 13 = 27$

Activity C starts after A → Starts at 14 → Finishes at $14 + 20 = 34$

Activity D starts after both B and C → Starts at $\max(27, 34) = 34$ → Finishes at $34 + 10 = 44$

QUESTION 7:

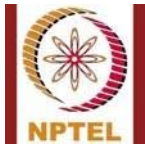
Which one of the following is true of a critical path in a PERT chart?

- a) It is the path having the maximum number of tasks.
- b) It is the path having the minimum number of tasks.
- c) It is the shortest path in terms of time
- d) It is the longest path in terms of time
- e) It is the path with the largest amount of slack time

Correct Answer: d) It is the longest path in terms of time

Detailed Solution:

In a PERT chart, the critical path is the longest duration path through the network, determining the minimum time required to complete the entire project.



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**QUESTION 8:**

Which one of the following statements regarding critical paths in a PERT chart is true?

- a) A critical path through a PERT chart is any path through the chart that contains the least number of edges.
- b) Some activities on the critical path can have slack.
- c) All activities on the critical path can have slack.
- d) Every PERT chart has exactly one critical path.
- e) It is possible that in the PERT chart for a project, there can be multiple critical paths, all having exactly the same duration.

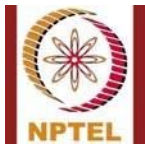
Correct Answer: e) It is possible that in the PERT chart for a project, there can be multiple critical paths, all having exactly the same duration.

Detailed Solution:

A PERT chart can have multiple critical paths if there are two or more paths with the same longest duration, each determining the project's completion time.

Answer questions 8 and 9 for a project whose activities, their precedence ordering, estimated time for completion are given in the following table.

Activity	Immediate follower	Estimated time in weeks
A	E,B	3
B	C,F	2
C	D,G	8
D	F	4
E	F	5
F	---	10
G	---	10



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QUESTION 9:

Which one of the following sequences of activities is on the critical path?

- a) A-E-F
- b) A-B-C-G
- c) A-B-C-D-F
- d) A-B-F
- e) C-D-G

Correct Answer: c) A-B-C-D-F

Detailed Solution:

For path $A \rightarrow B \rightarrow C \rightarrow D \rightarrow F$, Total duration is $\text{Total} = 3 + 2 + 8 + 4 + 10 = 27$. This is the maximum among the given options.

QUESTION 10:

Which one of the following paths has the greatest slack time?

- a) A-E-F
- b) A-B-C-G
- c) C-D-G
- d) A-B-C-D-F
- e) A-B-F

Correct Answer: e) A-B-F

Detailed Solution:

Slack = Critical path duration – Path duration

a) $A-E-F = 3 + 5 + 10 = 18 \rightarrow \text{Slack} = 27 - 18 = 9$

b) $A-B-C-G = 3 + 2 + 8 + 10 = 23 \rightarrow \text{Slack} = 27 - 23 = 4$

c) C-D-G = (invalid as C depends on B, and G on C)

d) $A-B-C-D-F = 27 \rightarrow \text{Slack} = 0$ (Critical path)

e) $A-B-F = 3 + 2 + 10 = 15 \rightarrow \text{Slack} = 27 - 15 = 12$ (This is the slack)
